

Rachit Sharma

Collider Phenomenology · Effective Field Theory · Flavour Physics · Collider–Cosmology Interface

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My research focuses on collider phenomenology of beyond-the-Standard-Model physics, with particular emphasis on leptoquark models, effective field theory, and the interface between collider and flavour observables. I derive UV-consistent constraints that simultaneously enforce LHC direct search limits, SMEFT operator correlations, and low-energy flavour data, exploiting the complementary information across experimental frontiers. My doctoral work at IISER Thiruvananthapuram established comprehensive LHC constraints on scalar and vector leptoquarks using a unified recasting framework, derived the complete gauge-invariant effective operator basis for vector LQs from first principles, and studied LQ-mediated RHN production including displaced-vertex signatures. I also work on extended scalar sector phenomenology, photon-initiated production, and simulation-based statistical inference. I am a developer of TOOLQIT, a public leptoquark recasting toolkit.

SELECTED RESEARCH HIGHLIGHTS

- Built a unified multi-channel χ^2 recasting framework for scalar and vector LQs, shifting exclusion boundaries by 20–40% relative to simplified-model analyses (*Phys. Rev. D* 109, 2024; *Phys. Rev. D* 113, 2026).
- Derived the complete gauge-invariant EFT operator basis for vector leptoquarks from first principles, enabling UV-consistent LHC limit reinterpretation (*Phys. Rev. D* 113, 2026).
- Established LHC constraints on RHN production via LQ mediators. (*Phys. Rev. D* 111, 2025).
- Developed TOOLQIT, an open-source toolkit for recasting LHC collider limits on leptoquark models.
- Applied BDT/DNN and simulation-based inference to EFT coefficient extraction and extended scalar sector searches at the HL-LHC.

EDUCATION

Ph.D. in Physics · IISER Thiruvananthapuram

Aug 2021 – Jun 2026

CGPA: 9.40/10 | Supervisor: Dr. Tanumoy Mandal | Thesis: *From Exclusion Bounds to Novel Signatures: A Comprehensive Study of Leptoquark Phenomenology*

M.Sc. in Physics · IIT Indore · CGPA: 8.32/10

Jul 2018 – Jul 2020

B.Sc. (Hons) in Physics · University of Delhi (DDU College) · 81.56%

Jul 2014 – Jul 2017

RESEARCH INTERESTS

- Leptoquark and BSM collider phenomenology:** scalar and vector leptoquarks at the LHC; long-lived particles and displaced signatures; vector-like fermions; extended scalar sectors.
- Effective field theory and SMEFT:** gauge-invariant operator derivation from first principles; UV-to-SMEFT matching; operator correlations at colliders; SMEFT global analyses and HL-LHC projections.
- Flavour physics and rare decays:** LQ-mediated contributions to $B \rightarrow K^{(*)} \nu \bar{\nu}$, $K \rightarrow \pi \nu \bar{\nu}$, $B \rightarrow D^{(*)} \tau \nu$, and $(g-2)_\mu$; correlated collider and flavour constraints on BSM parameter space.
- Collider–cosmology interface:** right-handed neutrino phenomenology and connections to leptogenesis; extended scalar sectors and electroweak phase transitions; gravitational-wave signatures and LISA forecasts.
- Statistical inference and ML methods:** simulation-based inference; DNN/BDT likelihood estimation; Bayesian model comparison; public recasting tools.

PUBLICATIONS

7 publications (5 published in peer-reviewed journals + 1 accepted at *Phys. Rev. D*; 1 under review at *Phys. Rev. Lett.*) · 1 conference proceedings · INSPIRE profile: inspirehep.net/authors/2739739

- A. Bhaskar, A. Das, T. Mandal, S. Mitra, **R. Sharma**, *Fresh look at the LHC limits for scalar leptoquarks*, **Phys. Rev. D** 109, 055018 (2024). doi:10.1103/PhysRevD.109.055018

Contribution: Built χ^2 recasting framework from scratch; all LHC channels; derived $(g-2)_\mu$ compatible regions.

- G. Duraikandan, R. Khanna, T. Mandal, S. Mitra, **R. Sharma**, *Right-handed neutrino production through first-generation leptoquarks*, **Phys. Rev. D** 111, 075032 (2025). doi:10.1103/PhysRevD.111.075032

Contribution: Led kinematic optimisation; studied displaced-vertex signatures of long-lived RHNs; derived LQ–Yukawa coupling constraints directly relevant to low-scale leptogenesis; mentored two M.Sc. students.

3. A. Das, T. Mandal, S. Mitra, **R. Sharma**, *Fresh look at the LHC limits on vector leptoquarks*, **Phys. Rev. D** **113**, 075022 (2026). doi:10.1103/4rqk-rjjr

Contribution: Full simulation chain; derived effective operators from gauge invariance and unitarity; co-wrote.

4. T. Mandal, S. Moretti, **R. Sharma**, *Non-standard decays of vector-like top partners in a 2HDM at the HL-LHC*, **Phys. Rev. D** (accepted 2026), arXiv:2606.20054. arxiv.org/abs/2606.20054

Contribution: BDT-optimised $2\tau + 2b + E_T^{\text{miss}}$ analysis; studied extended scalar sector relevant to EWPT and baryogenesis; co-wrote manuscript.

5. T. Mandal, S. Mitra, **R. Sharma**, *Photon-initiated enhancements in pair production of highly charged coloured particles*, arXiv:2605.21148 (2026). (Submitted to *Phys. Rev. Lett.*) arxiv.org/abs/2605.21148

Contribution: Led full analysis; photon PDF effects and EW corrections.

6. S. Dubey, N. Kumar, T. Mandal, S. Mitra, **R. Sharma**, *Vectorlike τ production through leptoquarks*, arXiv:2508.18047 (2025). (Reviewer report received, *Phys. Rev. D*) arxiv.org/abs/2508.18047

Contribution: Led kinematic variable study; long-lived particle signatures; mentored M.Sc. student.

7. A. Bhaskar, A. Das, T. Mandal, S. Mitra, **R. Sharma**, *TooLQit: Leptoquark Models and Limits*, arXiv:2412.19729 (2024). arxiv.org/abs/2412.19729

Contribution: Developer of the public toolkit; collider datasets and cross-section inputs.

CONFERENCE PROCEEDINGS

1. **R. Sharma**, T. Mandal, S. Mitra, *Improved Exclusion Limits for Leptoquarks and Associated BSM Searches at the HL-LHC*, **Springer Proc. Phys.** **432**, 288–291 (2026). Oral presentation, DAE–BRNS HEP Symposium (DAE–HEP 2024).

ONGOING PROJECTS

1. **R. Sharma**, T. Mandal, S. Mitra, J. C. Sam, *Prospects for trilepton final states in Left–Right Symmetric Models*. (In preparation.)

Contribution: Led collider analysis; LRSM gauge sector and W_R boson phenomenology; displaced-vertex and multi-lepton signatures of long-lived RHNs; signal–background optimisation.

2. **R. Sharma**, T. Mandal, S. Mitra, *EFT inference for third-generation leptoquark operators at the HL-LHC*. (Ongoing.)

Contribution: EFT coefficient extraction for 3rd-gen LQ + N_R signatures; Bayesian parameter inference using simulation-based methods.

CONFERENCES, TALKS & WORKSHOPS

Invited Seminar , Charles University, Prague [Invited talk]	2026
Extended Scalar Sectors at All Angles 2026 , CERN, Geneva [Oral]	2026
LHCp 2026 , Sorbonne University, Paris [Poster]	2026
Iwate Collider School (ICS 2025) , Iwate, Japan [Poster]	2025
DAE–BRNS HEP Symposium (DAE–HEP 2024) , IIT BHU [Oral]	2024
Phenomenology Symposium (PPC 2024) , IIT Hyderabad [Participant]	2024
ML4HEP Workshop , ICTS Bangalore [Workshop]	2023
DAE–BRNS HEP Symposium (DAE–HEP 2022) , IISER Mohali [Participant]	2022

AWARDS & FELLOWSHIPS

Prime Minister's Research Fellowship (PMRF) , Cycle 9 – competitive national fellowship; <150 awardees per cycle across all science disciplines	May 2022
Best Poster Award , Frontiers in Physics and Statistics (FPS-2024), IISER TVM	2024
NET JRF (All India Rank 149) & NET LS (AIR 64)	2020–21
GATE AIR 515 (2021), AIR 986 (2020) · JEST AIR 303 (2021) · IIT JAM AIR 426 (2018)	2018–21

TECHNICAL EXPERTISE

Programming	Python, C++, Mathematica, Bash
Simulation	MadGraph5, Pythia8, Delphes, CheckMATE, FeynRules, SARAH
Statistics/ML	DNNs, BDTs; simulation-based inference (MADMINER); Bayesian inference; TOOLQIT (developer)
EFT tools	WILSON, FLAVIO; SMEFT operator basis; χ^2 recasting frameworks
Symbolic calc.	Mathematica, FeynCalc, Package-X
HPC/Data	SLURM, ROOT, NumPy/SciPy, Git

TEACHING & MENTORING

- **Teaching Assistant (6 semesters)**: Electrodynamics & Special Relativity, Nuclear & Particle Physics, Electromagnetism, Mechanics.
- **Research mentoring**: Guided **four M.Sc. students** to completion, each resulting in a peer-reviewed publication; currently mentoring three further M.Sc. students.
- **Project supervision**: BS–MS thesis projects on spontaneous symmetry breaking and the Higgs mechanism.

ADDITIONAL INFORMATION

Languages	English (full professional), Hindi (native)
Citizenship	Indian
PhD awarded	June 2026
Availability	From December 2026 (or earlier if required)

REFERENCES

Dr. Tanumoy Mandal Associate Professor School of Physics IISER Thiruvananthapuram Kerala, India tanumoy@iisertvm.ac.in	Dr. Subhadip Mitra Associate Professor Dept. of Physics IIIT Hyderabad Telangana, India subhadip.mitra@iiit.ac.in	Prof. Stefano Moretti Professor Dept. of Physics & Astronomy Uppsala University Uppsala, Sweden s.moretti@physics.uu.se
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